

WAAM Digital Twin — Formula Reference

waam_twin project

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Abstract

Consolidated LaTeX formulae for the WAAM (Wire Arc Additive Manufacturing) digital twin study. Equations are grouped by physics domain and aligned with the `waam_twin` implementation (Taichi LBM + enthalpy-porosity + VOF), platform architecture, and validation references.

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1 Process inputs and energy balance

1.1 Electrical arc power

$$P = IV \quad (1)$$

$$Q_{\text{eff}} = \eta IV \quad (2)$$

where I is arc current [A], V arc voltage [V], and η arc thermal efficiency.

References: [25, 21, 35]

1.2 Wire mass feed

Wire cross-sectional area $A_w = \pi d_w^2/4$:

$$\dot{m} = \rho A_w v_{\text{wire}} = \rho \frac{\pi d_w^2}{4} v_{\text{wire}} \quad (3)$$

with wire diameter d_w [m], feed speed v_{wire} [m/s], density ρ [kg/m³].

References: [42, 28, 3]

1.3 Droplet mass and volume

For droplet frequency f_{drop} [Hz] and interval Δt_{drop} :

$$m_{\text{drop}} = \dot{m} \Delta t_{\text{drop}} = \frac{\dot{m}}{f_{\text{drop}}} \quad (4)$$

$$V_{\text{drop}} = \frac{m_{\text{drop}}}{\rho} \quad (5)$$

References: [42, 28, 3]

Equivalent spherical radius in grid cells (volume V , cell size Δx):

$$r_{\text{cells}} = \left(\frac{3V}{4\pi \Delta x^3} \right)^{1/3} \quad (6)$$

References: [42, 3]

1.4 Travel speed and heat input per unit length

$$q_{\text{line}} = \frac{Q_{\text{eff}}}{v_{\text{travel}}} \quad (7)$$

References: [37, 15, 25]

2 Governing conservation equations

2.1 Continuity

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0 \quad (8)$$

In the incompressible LBM liquid region, $\rho \approx \rho_0$.

References: [5, 26]

2.2 Momentum (Navier–Stokes)

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot [\mu(\nabla \mathbf{u} + \nabla \mathbf{u}^T)] + \rho \mathbf{g} + \mathbf{F}_{\text{body}} \quad (9)$$

Body-force contributions $\mathbf{F}_{\text{body}}$ include Marangoni, Lorentz, buoyancy, recoil, gas shear, and droplet impact (Sections 7–8).

References: [31, 25, 21]

2.3 Energy / enthalpy

Primary conserved variable is volumetric enthalpy H [J/m³]:

$$\frac{\partial H}{\partial t} + \mathbf{u} \cdot \nabla H = \nabla \cdot (k \nabla T) + \dot{q}_{\text{arc}} \quad (10)$$

with $k = \rho c_p \alpha$ and temperature T recovered from H (Section 4).

References: [41, 21, 7]

3 Heat sources

3.1 Gaussian surface flux

Two-dimensional Gaussian distribution on the pool surface (implementation in `inject_arc_heat`):

$$q(r) = \frac{Q \eta}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) \quad (11)$$

$$\Delta H_{ijk} = q(r) \frac{\Delta t}{\Delta x^3} \quad (12)$$

where r is distance from arc centre and σ is the Gaussian radius.

References: [38, 15, 3]

3.2 Goldak double-ellipsoid source

Classical Goldak (1984) form for front (f) and rear (r) ellipsoids:

$$q(\xi, y, z) = \frac{6\sqrt{3} f Q}{\pi\sqrt{\pi} a b c} \exp\left(-\frac{3\xi^2}{a^2} - \frac{3y^2}{b^2} - \frac{3z^2}{c^2}\right) \quad (13)$$

with partition fractions $f_f + f_r = 1$ and semi-axes (a, b, c) .

References: [15, 16]

Simplified 3D implementation (`inject_goldak_heat`): transverse radius σ , depth scales d_f, d_r , travel direction sign s :

$$r_{xy}^2 = (x - x_0)^2 + (y - y_0)^2 \quad (14)$$

$$\xi_{\text{eff}} = \frac{z - z_0}{d_{f/r}} \quad (15)$$

$$r^2 = r_{xy}^2 + 4\xi_{\text{eff}}^2 \quad (16)$$

$$q = \frac{Q \eta f_{f/r}}{2\pi\sigma^2} \exp\left(-\frac{r^2}{2\sigma^2}\right) \quad (17)$$

with $f_{f/r}$ selected by $(x - x_0) s \geq 0$.

References: [15, 16, 3]

3.3 Arc penetration attenuation

Below-surface weighting (exponential attenuation over penetration depth δ):

$$w_{\text{dep}}(z) = \exp\left(-\frac{\max(0, z_{\text{arc}} - z)}{\delta}\right) \quad (18)$$

References: [15, 37, 3]

3.4 Rosenthal validation (2D trailing edge)

Analytical far-field reference (`validation/rosenthal.py`):

$$\sigma_R = \frac{\rho c_p v}{2k} \quad (19)$$

$$T - T_0 = \frac{\eta Q}{2\pi k} \frac{1}{|x|} \exp(\sigma_R x), \quad x < 0 \quad (20)$$

References: [37, 3]

4 Thermal transport and phase change

4.1 Fourier conduction

$$\mathbf{q} = -k \nabla T, \quad \alpha = \frac{k}{\rho c_p} \quad (21)$$

References: [21, 9]

4.2 Enthalpy–porosity method

Define

$$H_{\text{sol}} = \rho c_p T_s \quad (22)$$

$$H_{\text{liq}} = H_{\text{sol}} + \rho L_f \quad (23)$$

where T_s , T_l are solidus and liquidus temperatures, L_f specific latent heat.

Solid ($H \leq H_{\text{sol}}$):

$$f_l = 0, \quad T = \frac{H}{\rho c_p} \quad (24)$$

Mushy zone ($H_{\text{sol}} < H < H_{\text{liq}}$):

$$f_l = \frac{H - H_{\text{sol}}}{\rho L_f} \quad (25)$$

$$T = T_s + f_l (T_l - T_s) \quad (26)$$

Liquid ($H \geq H_{\text{liq}}$):

$$f_l = 1, \quad T = T_l + \frac{H - H_{\text{liq}}}{\rho c_p} \quad (27)$$

References: [41, 7, 3]

4.3 Lattice thermal update

In lattice units ($\alpha_{\text{lu}} = \alpha \Delta t / \Delta x^2$):

$$\left. \frac{\partial H}{\partial t} \right|_{\text{cell}} = -\mathbf{u} \cdot \nabla H + \rho c_p \alpha_{\text{lu}} \nabla^2 T \quad (28)$$

(first-order upwind advection, central diffusion).

References: [21, 32, 3]

4.4 Enthalpy ceiling (vaporization cap)

$$H \leq H_{\text{vap}} = \rho c_p T_{\text{vap,cap}} + \rho L_{\text{vap}} \quad (29)$$

References: [29, 3]

4.5 Stefan solidification benchmark

Stefan number and interface position (`test_stefan_solidification`):

$$\text{Ste} = \frac{c_p (T_{\text{init}} - T_0)}{L_f} \quad (30)$$

$$\sqrt{\pi} \lambda \exp(\lambda^2) \text{erf}(\lambda) = \text{Ste} \quad (31)$$

$$s(t) = 2\lambda\sqrt{\alpha t} \quad (32)$$

References: [9, 11, 3]

4.6 Boundary heat losses

On gas-exposed cells:

$$q_{\text{loss}} = h_{\text{conv}} (T - T_{\text{amb}}) + \varepsilon \sigma_{\text{SB}} (T^4 - T_{\text{amb}}^4) \quad (33)$$

$$H \leftarrow H - q_{\text{loss}} \frac{\Delta t}{\Delta x} \rho c_p \quad (34)$$

References: [21, 25, 3]

5 Lattice Boltzmann numerics

5.1 Lattice units

$$\Delta t = \Delta x \frac{u_{\text{ref,lu}}}{u_{\text{ref,phys}}} \quad (35)$$

$$\nu_{\text{lu}} = \frac{\mu}{\rho} \frac{\Delta t}{\Delta x^2}, \quad \tau = 3\nu_{\text{lu}} + \frac{1}{2} \quad (36)$$

$$\alpha_{\text{lu}} = \alpha \frac{\Delta t}{\Delta x^2}, \quad \tau_T = 3\alpha_{\text{lu}} + \frac{1}{2} \quad (37)$$

Target Mach number $\text{Ma} = u_{\text{ref,phys}} \Delta t / \Delta x \approx 0.05$.

References: [26, 10, 3]

5.2 D3Q19 equilibrium (SRT)

$$f_i^{\text{eq}} = w_i \rho \left(1 + 3 \mathbf{e}_i \cdot \mathbf{u} + \frac{9}{2} (\mathbf{e}_i \cdot \mathbf{u})^2 - \frac{3}{2} \mathbf{u}^2 \right) \quad (38)$$

References: [34, 26, 10]

5.3 Guo forcing scheme (2002)

Velocity shift and forcing source:

$$\mathbf{u}^* = \mathbf{u} + \frac{\tau}{\rho} \mathbf{F} \quad (39)$$

$$S_i = w_i \left(1 - \frac{\omega}{2}\right) [3 \mathbf{e}_i \cdot \mathbf{F} + 9(\mathbf{e}_i \cdot \mathbf{u})(\mathbf{e}_i \cdot \mathbf{F}) - 3 \mathbf{u} \cdot \mathbf{F}] \quad (40)$$

Lattice force scaling: $F_{\text{lu}} = F_{\text{phys}} \Delta t^2 / (\rho \Delta x)$.

References: [19, 26, 3]

5.4 Carman–Kozeny mushy-zone drag

Semi-implicit velocity correction before collision:

$$\mathbf{u}_{\text{final}} = \frac{\mathbf{u}^*}{1 + C_K \frac{(1 - f_l)^2}{f_l^3 + \varepsilon}} \quad (41)$$

References: [41, 33, 3]

6 Free surface (VOF) and surface tension

6.1 VOF phase field

Volume fraction $\phi \in [0, 1]$; interface where $|\nabla \phi| > \varepsilon$.

References: [20, 36]

6.2 Continuum Surface Force (CSF)

Balanced-force surface tension:

$$\hat{\mathbf{n}} = \frac{\nabla \phi}{|\nabla \phi|} \quad (42)$$

$$\kappa = -\nabla \cdot \hat{\mathbf{n}} \quad (43)$$

$$\mathbf{F}_{\text{CSF}} = \gamma \kappa \nabla \phi \quad (44)$$

References: [8, 3]

6.3 Marangoni force

Surface-tension gradient driven flow ($d\gamma/dT < 0$ for pure metals $\Rightarrow$ outward flow):

$$\nabla_s T = (\mathbf{I} - \hat{\mathbf{n}} \hat{\mathbf{n}}^\top) \nabla T \quad (45)$$

$$\mathbf{F}_M = \frac{d\gamma}{dT} \nabla_s T |\nabla \phi| \quad (46)$$

Surface shear stress form (FEM boundary analogue):

$$\tau_M = \frac{d\gamma}{dT} \nabla_s T \quad (47)$$

References: [25, 12, 31, 3]

6.4 Wetting and contact angle

Young's law:

$$\cos \theta = \frac{\gamma_{sv} - \gamma_{sl}}{\gamma_{lv}} \quad (48)$$

References: [13, 8, 14]

Effective interface normal at wall:

$$\hat{\mathbf{n}}_{\text{eff}} = \sin \theta \hat{\mathbf{n}}_{\text{wall}} + \cos \theta \hat{\mathbf{t}}_{\text{wall}} \quad (49)$$

Wall capillary curvature approximation:

$$\kappa_{\text{wall}} \approx \frac{2 \cos \theta}{\Delta x} \quad (50)$$

References: [8, 14, 2, 3]

Capillary length:

$$\lambda_c = \sqrt{\frac{\gamma}{\rho g}} \quad (51)$$

References: [13, 14, 8, 3]

6.5 Bond number

$$\text{Bo} = \frac{\rho g L^2}{\gamma} \quad (52)$$

References: [5, 13]

For steel-scale $L \sim 5$ mm, $\text{Bo} \sim 0.1$: gravity is secondary to surface tension but affects crown height.

References: [5, 3]

7 Weld-pool body forces

7.1 Lorentz (electromagnetic) force

$$\mathbf{F}_{\text{em}} = \mathbf{J} \times \mathbf{B} \quad (53)$$

Electrical potential (conductive pool, σ electrical conductivity):

$$\nabla \cdot (\sigma \nabla \phi_e) = 0 \quad (54)$$

Current density (Ohm's law):

$$\mathbf{J} = -\sigma \nabla \phi_e \quad (55)$$

Magnetic field (Ampère's law, magnetostatic approximation):

$$B_x = \mu_0 \left(\frac{\partial J_y}{\partial z} - \frac{\partial J_z}{\partial y} \right) \quad (56)$$

$$B_y = \mu_0 \left(\frac{\partial J_z}{\partial x} - \frac{\partial J_x}{\partial z} \right) \quad (57)$$

$$B_z = \mu_0 \left(\frac{\partial J_x}{\partial y} - \frac{\partial J_y}{\partial x} \right) \quad (58)$$

with $\mu_0 = 4\pi \times 10^{-7}$ H/m.

References: [31, 23, 18, 3]

7.2 Boussinesq thermal buoyancy

$$F_z = -\rho g \beta (T - T_{\text{ref}}) f_l \quad (59)$$

where β is volumetric thermal expansion coefficient.

References: [31, 25, 21]

7.3 Hydrostatic gravity on liquid

$$\mathbf{F}_g = \rho \mathbf{g} f_l, \quad \mathbf{g} = (0, 0, -g) \quad (60)$$

Additive to Boussinesq term (59).

References: [3, 2]

7.4 Vapor recoil pressure

Constant model:

$$p_{\text{recoil}} = p_0 \cdot f(T/T_{\text{ref}}) \quad (61)$$

applied as downward surface pressure on the free surface.

References: [29, 3]

Clausius–Clapeyron model (`apply_vapor_recoil_clausius_clapeyron`):

$$P_{\text{vap}}(T) = P_{\text{ref}} \exp \left[\frac{L_v}{R_v} \left(\frac{1}{T_b} - \frac{1}{T} \right) \right], \quad T > T_b \quad (62)$$

where L_v is latent heat of vaporization, R_v specific gas constant, T_b boiling temperature.

References: [29, 39, 3]

7.5 Gas-jet shear stress

Dynamic shear scale:

$$\tau_{\text{gas}} = \frac{1}{2} \rho_{\text{gas}} v_{\text{jet}}^2 C_\tau \quad (63)$$

applied tangentially on the pool surface in the radial direction from the arc centre.

References: [30, 31, 3]

7.6 Arc pressure

Downward Gaussian pressure distribution on the free surface (magnitude p_{arc}):

$$p_{\text{arc}}(r) = p_0 \exp \left(-\frac{r^2}{2\sigma^2} \right) \quad (64)$$

References: [31, 40, 3]

8 Droplet deposition and impact

8.1 Wire mass balance

Expected deposited mass over simulation time t :

$$m_{\text{dep,expected}} = \dot{m} t \quad (65)$$

References: [3, 42]

Mass balance ratio (validation):

$$\chi_m = \frac{m_{\text{dep,sim}}}{m_{\text{dep,expected}}} \approx 1 \quad (66)$$

References: [3, 42]

8.2 Droplet impact velocity

Combined feed, gravity, and pinch terms (`droplet_impact_velocity_m_s`):

$$L_{\text{drop}} \approx \frac{v_{\text{wire}}}{f_{\text{drop}}} \quad (67)$$

$$v_{\text{grav}} = \sqrt{2gL_{\text{drop}}} \quad (68)$$

$$v_{\text{impact}} \approx f_{\text{mode}} \max(v_{\text{wire}}, v_{\text{grav}} + v_{\text{pinch}}) \quad (69)$$

with transfer-mode scale $f_{\text{mode}} \in \{0.78, 1.05, 1.18\}$ for globular / pulsed / spray.

References: [42, 28, 3]

8.3 Droplet impact pressure (kinetic)

$$E_{\text{kin}} = \frac{1}{2} m_{\text{drop}} v_{\text{impact}}^2 \quad (70)$$

$$p_{\text{impact}} \approx \frac{E_{\text{kin}}}{A_{\text{contact}}} = \frac{\frac{1}{2} m_{\text{drop}} v_{\text{impact}}^2}{\pi r_{\text{contact}}^2} \quad (71)$$

References: [42, 6, 3]

8.4 Deposition footprint

Footprint radius for surface-limited deposition:

$$r_{\text{foot}} = \max\left(r_{\text{drop}}, \frac{w_{\text{pool}}}{2}\right) \quad (72)$$

References: [2, 3]

8.5 Droplet entry enthalpy

$$H_{\text{drop}} = \rho c_p T_{\text{drop}} + \rho L_f \quad (73)$$

with $T_{\text{drop}} = T_l + \Delta T_{\text{superheat}}$.

References: [42, 41, 3]

9 Electrical stick-out and CTWD

9.1 Stick-out resistance

$$R_{\text{stick}} = \frac{\rho_e L_{\text{stick}}}{A_w} \quad (74)$$

$$P_{\text{preheat}} = \eta_{\text{stick}} I^2 R_{\text{stick}} \quad (75)$$

References: [24, 21, 2]

Wire preheat temperature rise:

$$\Delta T_{\text{preheat}} = \frac{P_{\text{preheat}}}{\dot{m} c_p} \quad (76)$$

References: [24, 2, 3]

9.2 CTWD update (layer drift)

$$\text{CTWD}_{n+1} = \text{CTWD}_{\text{nominal}} + (h_{\text{bead,measured}} - h_{\text{layer,planned}}) \quad (77)$$

References: [24, 2, 3]

10 Material properties and constitutive relations

$$c_{p,\rho} = \rho c_p \quad [\text{J}/(\text{m}^3 \text{K})] \quad (78)$$

$$L_\rho = \rho L_f \quad [\text{J}/\text{m}^3] \quad (79)$$

$$\gamma(T) \approx \gamma_0 + \frac{d\gamma}{dT} (T - T_{\text{ref}}) \quad (80)$$

$$\mu(T), k(T), c_p(T) \quad \text{piecewise-linear tables in material YAML} \quad (81)$$

References: [21, 25, 3]

Surface tension temperature coefficient $d\gamma/dT$ drives Marangoni; sign change with surface-active elements (S, O) flips pool circulation direction.

References: [25, 30, 3]

11 Bead geometry metrics

$$w_{\text{pool}} = \text{transverse extent of liquid region at surface} \quad (82)$$

$$d_{\text{pool}} = \text{penetration depth below substrate reference} \quad (83)$$

$$h_{\text{bead}} = \max z \text{ of deposited solid above substrate} \quad (84)$$

$$\theta_{\text{toe}} \approx \text{contact angle at solid-liquid-gas triple line} \quad (85)$$

Aspect ratio and validation:

$$\frac{h}{w}, \quad \frac{d}{w} \quad (86)$$

References: [3, 25, 2]

12 Coordinate frames (robot integration)

KUKA TCP [mm] to simulation metres (`kuka_adapter.tcp_mm_to_sim_m`):

$$x_{\text{sim}} = \frac{x_{\text{TCP,mm}} - x_{\text{origin,mm}}}{1000} - x_{\text{offset,m}} \quad (87)$$

$$y_{\text{sim}} = \frac{y_{\text{TCP,mm}} - y_{\text{origin,mm}}}{1000} - y_{\text{offset,m}} \quad (88)$$

$$z_{\text{sim}} = \frac{z_{\text{TCP,mm}} - z_{\text{origin,mm}}}{1000} \quad (89)$$

References: [3, 27]

13 Digital twin platform (architecture signals)

13.1 Twin input snapshot

$$\mathcal{U}(t) = \{t, x, y, z, I, V, \text{WFS}, \text{is_welding}, \text{material_id}\} \quad (90)$$

References: [1, 22, 17]

13.2 Twin step

$$\mathcal{S}(t + \Delta t) = \text{WAAMTwin.step}(\mathcal{U}(t)) \quad (91)$$

State outputs: temperature field T , liquid fraction f_l , VOF ϕ , bead metrics (w, h, d) , telemetry JSON.

References: [3, 1, 17]

13.3 Supervisory closed-loop deviation

$$\Delta y = y_{\text{meas}} - y_{\text{twin}}, \quad y \in \{I, V, T_{\text{pool}}, W, H\} \quad (92)$$

References: [1, 24, 4]

Bounded setpoint correction (example):

$$I_{\text{new}} = \text{clip}(I + K_I \Delta I, I_{\text{min}}, I_{\text{max}}) \quad (93)$$

References: [24, 1, 4]

13.4 Synchronization

Pose sample index n aligned to sensor timestamp t_n :

$$t_n = t_0 + n \Delta t_{\text{sample}}, \quad \mathbf{x}_{\text{TCP}}(t_n) \leftrightarrow \mathcal{U}_n \quad (94)$$

References: [1, 22, 17]

14 Dimensionless groups (reference)

$$\text{Re} = \frac{\rho U L}{\mu} \quad \text{Pr} = \frac{\nu}{\alpha} \quad (95)$$

$$\text{Pe} = \frac{U L}{\alpha} \quad \text{Ma} = \frac{U c_s}{a_{\text{sound}}} \approx \frac{u_{\text{ref}} \Delta t}{\Delta x} \quad (96)$$

$$\text{We} = \frac{\rho U^2 L}{\gamma} \quad \text{Bo} = \frac{\rho g L^2}{\gamma} \quad (97)$$

References: [5, 21, 26]

15 Coupled timestep order (reference)

Per-step operator sequence in `solvers/coupled_step.py`:

1. Arc heat injection $\rightarrow$ wire deposition $\rightarrow$ droplet impact
2. Thermal advection–diffusion; phase change; optional bead freeze
3. VOF advection, reinit, contact-angle BC, flag update
4. CSF, Marangoni, hydrostatic gravity, Boussinesq buoyancy
5. Lorentz, gas shear, arc pressure, recoil
6. LBM collide–stream; tracer advection

References: [3, 41, 8, 19]

16 Bibliography

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